

5MW Wind Turbine Control System Implementation

This repository contains a MATLAB/Simulink environment for modeling and controlling a 5 MW Floating Offshore Wind Turbine (FOWT). The model focuses on Region 3 operation, where the primary control objective is to maintain rated power and generator speed through active pitch control.

Model Architecture

The simulation is divided into three primary functional blocks:

- **Wind Input:** A selector block allowing for constant wind (18 m/s), sinusoidal variations, or turbulent wind profiles generated via QBlade.
- **Controller Choice:** A modular section containing four distinct non-linear control laws for evaluation.
- **Aerodynamic Plant:** A simplified Matlab Function block representation of the turbine dynamics (Plant_Q1_Simple) used to test controller response and stability.

Control Strategies

The system allows the user to switch between four advanced control algorithms. Each offers specific advantages in handling the non-linearities of wind energy systems:

Controller	Description	Key Advantage
Input-Output Linearization	A technique that algebraically transforms non-linear system dynamics into linear ones.	Simplification: Allows the use of well-known linear control tools on a complex non-linear plant.
Sliding Mode Controller (SMC)	A robust control method that forces the system state onto a predefined "sliding surface".	Robustness: Exceptional at handling parameter uncertainties and external disturbances in wind speed.
Dynamic SMC	An evolution of SMC that incorporates higher-order derivatives into the control law	Smoothness: Reduces the "chattering" effect common in standard SMC, protecting the pitch actuators from wear.
Super-Twisting SMC	A specific type of Second-Order Sliding Mode (SOSM) algorithm.	Precision & Chattering Avoidance: Provides high-accuracy tracking without the high-frequency switching of traditional SMC.

Usage

1. Open and run the Wind_s.m file
2. Open the .slx file in MATLAB/Simulink.
3. Select the desired wind profile in the Wind Input block.
4. Choose one of the four controllers in the Controller Choice block.
5. Run the simulation and monitor out.Omega (Generator Speed) and out.Power to evaluate performance.